

For **DATASET-NAME = MIMIC-III**, the clinically appropriate causal level is not the raw laboratory value, vital sign, medication row, or charted item. The causal level should be **latent physiology**: chronic vulnerability, acute inflammatory/infectious stress, shock, respiratory failure, renal dysfunction, neurologic dysfunction, metabolic derangement, hepatic/coagulation dysfunction, cardiac strain, and global acute severity.

The strongest high-level causal pattern is:

background/case-mix → latent acute physiology and organ dysfunction → observed measurements, care intensity, measurement intensity → mortality.

Observed measurements in DATASET-NAME are mostly **manifestations or proxies** of hidden patient states. For example, creatinine does not “cause” mortality in the causal model; renal dysfunction contributes to mortality and generates abnormal creatinine, BUN, urine output, potassium, and acid-base findings. DATASET-NAME’s official documentation states that the database contains ICU demographics, vital signs, labs, medications, procedures, notes, imaging reports, and mortality, collected during routine care rather than by a fixed research protocol. This makes irregular sampling and informative missingness central to the causal representation. [physionet.org](#) +1

The main identifiability limitations are that many variables are dictionary-coded ITEMIDs, clinical measurements are ordered by clinicians, treatment decisions are confounded by severity, and chronic baseline state is only partially observed through demographics, ICD-9 codes, prior diagnoses, and admission context. The resulting DAG is therefore a **clinically plausible structural causal model**, not a proven ground-truth graph.

1. Dataset audit

DATASET-NAME is a relational ICU database. The public documentation describes 26 core tables linked by SUBJECT_ID , HADM_ID , and ICUSTAY_ID ; schema metadata also lists 40 physical tables because CHARTEVENTS is partitioned. Public table-level documentation is complete, but the full clinical variable inventory must be enumerated locally from D_ITEMS and D_LABITEMS , because many measurements are ITEMID-coded rather than exposed as flat columns. [physionet.org](#) +1

1.1 Core identifiers, background, and outcomes

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
SUBJECT_ID	Identifier	Patient-level identifier	Integer	Links patient across admissions.
HADM_ID	Identifier	Hospital admission identifier	Integer	Links admission-level events.
ICUSTAY_ID	Identifier	ICU stay identifier	Integer	Links ICU-specific events.
PATIENTS.GENDER	Background	Recorded sex/gender	Categorical	Background cause of risk and disease distribution.
PATIENTS.DOB	Background	Date of birth	Shifted timestamp	Used to derive age; age >89 is deidentified as very old.
PATIENTS.DOD , DOD_HOSP , DOD_SSN	Outcome / survival metadata	Date of death	Shifted timestamp	pod is null if alive at least 90 days post-discharge. lcp.mit.edu
PATIENTS.EXPIRE_FLAG	Outcome	Indicates patient died	Binary	Not allowed in latent-tag construction.
ADMISSIONS.ADMITTIME , DISCHTIME	Background / timing	Hospital admission and discharge	Timestamp	Used for cohort timing, not physiology.
ADMISSIONS.DEATHTIME	Outcome	Time of death	Timestamp	Outcome only; leakage if used for latent states.
ADMISSIONS.ADMISSION_TYPE	Background / care context	Emergency, elective, urgent, etc.	Categorical	Case-mix and confounding variable.
ADMISSIONS.ADMISSION_LOCATION	Background / care context	Origin before admission	Categorical	Proxy for illness pathway.
ADMISSIONS.DISCHARGE_LOCATION	Outcome-adjacent	Disposition	Categorical	Do not use for latent physiology because it is post-baseline.

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
ADMISSIONS.INSURANCE , LANGUAGE , RELIGION , MARITAL_STATUS , ETHNICITY	Background / social-context proxies	Sociodemographic and access/context variables	Categorical	Possible confounders for care intensity, measurement intensity, and outcomes.
ADMISSIONS.DIAGNOSIS	Background / admission-context proxy	Free-text admitting diagnosis	Text	Useful but noisy; avoid direct outcome leakage.
ADMISSIONS.HOSPITAL_EXPIRE_FLAG	Outcome	In-hospital mortality	Binary	Outcome only.
ADMISSIONS.HAS_CHARTEVENTS_DATA	Measurement-process	Whether charted observations exist	Binary	Observation availability indicator. lcp.mit.edu
ICUSTAYS.FIRST_CAREUNIT , LAST_CAREUNIT	Background / care context	ICU type	Categorical	Case-mix and care-process determinant.
ICUSTAYS.INTIME , OUTTIME , LOS	Timing / outcome-adjacent	ICU entry, exit, length of stay	Timestamp / days	LOS is outcome-adjacent; avoid for early latent labels. lcp.mit.edu

1.2 Time-varying observed measurements

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
CHARTEVENTS	Observed time-varying / measurement process	Bedside charted measurements, vitals, ventilator settings, nursing observations, GCS, RASS, code status, some labs	ITEMID-coded, VALUE , VALUENUM , VALUEUOM	Very large event manifestation clinician docu
Heart rate	Observed time-varying	Cardiovascular stress, shock, fever, pain, arrhythmia	beats/min	Proxy for shock strain, global
Systolic/diastolic/MAP blood pressure	Observed time-varying	Perfusion pressure	mmHg	Proxy for shock
Respiratory rate	Observed time-varying	Respiratory distress / metabolic compensation	breaths/min	Proxy for resp metabolic der
Oxygen saturation	Observed time-varying	Peripheral oxygenation	%	Proxy for resp modified.
Temperature	Observed time-varying	Fever/hypothermia	°C or °F depending item	Proxy for infla normalization
Glasgow Coma Scale	Observed time-varying	Consciousness / neurologic function	3–15	Proxy for neu confounded k
Ventilator settings, FiO2, PEEP, ventilation status	Treatment / observed support	Respiratory support and oxygen delivery	Fraction, cmH2O, categorical	Care-process
LABEVENTS	Observed time-varying	Laboratory test results	ITEMID-coded, VALUE , VALUENUM , VALUEUOM , abnormal FLAG	Lab values are informative w
CBC: WBC, hemoglobin, platelets	Observed lab proxy	Inflammation, anemia, coagulation/hematologic dysfunction	K/μL, g/dL	Platelets are u dysfunction.
Chemistry: sodium, potassium, bicarbonate, chloride, BUN, creatinine, glucose	Observed lab proxy	Renal, metabolic, electrolyte, endocrine stress	mmol/L, mg/dL	Requires unit

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
Blood gas: pH, PaO ₂ , PaCO ₂ , base excess	Observed lab proxy	Respiratory and metabolic acid-base state	pH units, mmHg, mmol/L	Supports resp latent states.
Lactate	Observed lab proxy	Tissue hypoperfusion / cellular stress	mmol/L	Proxy for shock severity.
Bilirubin, AST, ALT, albumin	Observed lab proxy	Hepatic injury/synthetic function	mg/dL, U/L, g/dL	Hepatic dysfunction measured.
INR, PT, PTT	Observed lab proxy	Coagulation dysfunction	Ratio / seconds	Careful: anticoagulation possible.
Troponin, CK-MB	Observed lab proxy	Cardiac injury/strain	ng/mL, µg/L, assay-dependent	Sparse and often dependent

1.3 Interventions, care-process, and procedures

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
INPUTEVENTS_CV , INPUTEVENTS_MV	Treatment / care-process	Fluids, IV medications, infusions, nutrition	ITEMID-coded amounts/rates	Split by source system; fluid-input structure differs between CareVue and MetaVision. physionet.org +1
Vasopressor infusions	Treatment / care-process	Hemodynamic support	drug rate, dose units vary	Strong proxy for shock but also treatment variable.
IV fluids	Treatment / care-process	Resuscitation / volume management	mL, rates	Confounded by indication.
Antibiotics	Treatment / care-process	Suspected/treated infection	prescription rows	Proxy for clinician suspicion; not identical to infection.
PRESCRIPTIONS	Treatment / care-process	Ordered medications	drug, dose, route, dates	Use cautiously; time precision may vary.
PROCEDUREEVENTS_MV	Treatment / care-process	Timed procedures such as ventilation, dialysis, imaging	ITEMID-coded start/end	MetaVision only; includes procedure timing.
PROCEDURES_ICD , CPTEVENTS	Procedure / coding	Billing/procedure codes	ICD-9/CPT	Often post hoc and not always baseline-timed.
OUTPUTEVENTS	Observed time-varying / care-process	Urine, drains, fluid output	usually mL	Urine output is renal/perfusion proxy; output charting is care-process dependent.

1.4 Diagnoses, microbiology, notes, and dictionaries

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
DIAGNOSES_ICD	Background / phenotype metadata	ICD-9 diagnosis codes	ICD-9-CM	Useful for chronic burden, baseline comorbidity, and admission phenotype; no timestamps.
D_ICD_DIAGNOSES	Dictionary	Diagnosis code definitions	ICD-9 descriptions	Crosswalk for diagnosis concepts.
MICROBIOLOGYEVENTS	Observed proxy / care-process	Cultures, organisms, sensitivities	Specimen, organism, antibiotic interpretation	Negative cultures may be recorded by null organism name. physionet.org

Variable / table	Variable type	Clinical meaning	Unit or coding	Notes
NOTEEVENTS	Observed text / care-process	Notes, ECG reports, radiology reports, discharge summaries	Free text	Rich but high leakage risk if discharge summaries are used.
D_ITEMS	Dictionary	Non-lab charted item definitions	ITEMID dictionary	Required for variable extraction.
D_LABITEMS	Dictionary	Lab item definitions, LOINC where available	ITEMID dictionary	Required for lab extraction.
SERVICES , TRANSFERS	Background / care-process	Service line and location transitions	Categorical + timestamps	ICU type, service changes, and transfer patterns affect measurement and outcomes.
CALLOUT	Outcome-adjacent care-process	Readiness for transfer/discharge	Categorical/timestamps	Avoid for early mortality/physiology labels.

2. Candidate latent variables

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
Chronic baseline burden	Pre-ICU vulnerability from age, comorbidity, chronic organ disease, malignancy, frailty proxies	Age, ICD-9 diagnoses, admission type, ICU type, prior admissions if derived	High	Keep	Strong confounder for measurement of mortality.
Acute inflammation / infection burden	Dysregulated host response or suspected infection	Temperature, WBC, lactate, cultures, antibiotics, microbiology	Medium-high	Keep	Central ICU lab is partly unobserved, emphasizes infection dysfunction rather than PMC
Global acute severity	Multi-domain physiologic instability	Vitals, GCS, urine output, ventilation, labs	High	Keep	Severity score represents organ physiology rather than MSD Manuals +1
Circulatory shock	Inadequate perfusion / hemodynamic instability	MAP/SBP, lactate, vasopressors, urine output, pH/base deficit	High	Keep	Strong mortality, vasopressors must be separated
Respiratory failure	Hypoxemia, ventilatory failure, need for support	SpO2, PaO2/FiO2, FiO2, PEEP, ventilation, RR, PaCO2/pH	High	Keep	Strongly supports blood gases, ARDS threshold based severity
Renal dysfunction	AKI or clinically important renal failure	Creatinine, BUN, urine output, potassium, bicarbonate, dialysis	High	Keep	KDIGO support output criteria, outputs. MSC
Hepatic dysfunction	Liver injury or impaired clearance/synthetic function	Bilirubin, AST, ALT, albumin, INR	Medium	Merge with coagulation	Hepatic and coagulation are clinically related, measured sequentially.
Coagulation / hematologic dysfunction	Platelet and clotting-system derangement	Platelets, INR, PT, PTT, hemoglobin	Medium	Merge with hepatic	SOFA uses platelet organ domain implementation
Neurologic dysfunction	Depressed consciousness or acute brain dysfunction	GCS, RASS, pupil response,	Medium-high	Keep	GCS is widely used for sedation/intubation interpretation

Candidate latent variable	Clinical meaning	Supporting observed variables	Dataset support	Decision	Rationale
		sedation/intubation indicators			
Metabolic / acid-base-electrolyte derangement	Acidosis/alkalosis, lactate, electrolyte, glucose instability	pH, bicarbonate, lactate, base excess, Na, K, glucose	High	Keep	Cross-organ s mortality pat
Cardiac injury / strain	Myocardial injury, arrhythmia burden, cardiac stress	Troponin, CK-MB, HR, BP, ECG report, care unit	Medium-low	Keep, low confidence	Clinically imp sparse/selecti weak latent p
Volume status / fluid balance burden	Overload or depletion	Inputs, outputs, weight, edema notes, CVP	Low-medium	Reject as standalone	Measurement source system dependent.
Nutritional burden	Malnutrition/catabolism	Albumin, weight, nutrition inputs	Low	Reject	Albumin is nc inflammation,
Treatment intensity	Clinician response and organ support	Vasopressors, ventilation, dialysis, fluids, antibiotics, monitoring frequency	High	Do not treat as biological latent	Represent as primary laten
Missingness / observation intensity	Clinician-driven measurement process	Lab counts, chart counts, missing indicators	High	Separate measurement-process node	Informative rr collapsed intc

3. Final approved latent variables

le	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness str
_BURDEN	Baseline vulnerability before ICU physiology	Comorbidity/frailty is not directly measured	Age, ICD-9 comorbidity groups, chronic organ disease, malignancy, admission context	Score ≥2 among age, comorbidity, chronic organ disease, malignancy/immunosuppression, nonelective high-risk admission	Missing ICD → unknown/0 with not infer absent from missing co strongly
ATION_SEPSIS	Suspected infection/inflammatory burden	True infection and dysregulated host response are hidden	Temp, WBC, lactate, cultures, antibiotics, microbiology	Score ≥2, with at least one infection-evidence or inflammatory-evidence domain	Missing cultures not equal no inf include measure flag
SEVERITY	Multi-system acute physiologic severity	Severity is a construct across organs	Vitals, GCS, ventilation, urine output, lactate, labs	≥3 abnormal organ domains or ≥2 severe domains	Calculate from a domains; require minimum obser domains or mar uncertain
	Circulatory failure / tissue hypoperfusion	Perfusion adequacy is not directly measured	MAP/SBP, vasopressors, lactate, urine output, acidosis	Score ≥2 or vasopressor + hypoperfusion evidence	If BP missing bu vasopressor pre not mark missin normal
TORY_FAILURE	Hypoxemic or ventilatory failure	Lung gas exchange state is latent	SpO2, PaO2/FiO2, FiO2, PEEP, ventilation, RR, PaCO2/pH	Score ≥2 or mechanical ventilation with oxygenation/gas-exchange abnormality	Missing ABG alk SpO2/FiO2 fallb available
/SFUNCTION	AKI or clinically important renal failure	Kidney function is inferred from filtration/output	Creatinine, BUN, urine output, K, bicarbonate, dialysis	Score ≥2 or dialysis/procedure evidence	Require time wir missing urine or normal unless o table available
_COAG_DYSFUNC	Liver/coagulation/hematologic organ dysfunction	Synthetic/clearance state is not directly observed	Bilirubin, platelets, INR/PT/PTT, AST/ALT, albumin	Score ≥2	Missing INR/bili common; do no normal without

le	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness str
IC_DYSFUNCTI	Depressed consciousness/acute neurologic dysfunction	Neurologic state is partly obscured by sedation/intubation	GCS, RASS, pupil response, charted mental status	GCS abnormal plus another neuro/sedation-context indicator, or very low GCS alone	Missing GCS not sedation/intuba required
IC_DERANGEMEN	Acid-base, lactate, electrolyte, glucose instability	Systemic metabolic homeostasis is latent	pH, bicarbonate, base excess, lactate, Na, K, glucose	Score ≥2	If blood gas mis on chemistry; in count flag
_STRAIN	Myocardial injury/strain or clinically important arrhythmia burden	Cardiac injury is selectively tested and incompletely observed	Troponin, CK-MB, HR extremes, arrhythmia charting, ECG reports	Score ≥2, with biomarker or rhythm evidence required	Missing troponi unmeasured, no low-confidence

4. Rule-based binary decision trees

All rules use an early analysis window, preferably **first 24 or first 48 ICU hours**, with no mortality, discharge location, survival time, ICU length of stay, or post-outcome data. Use `first`, `worst`, `min`, `max`, `mean`, `slope`, `count`, and `missing` summaries computed only within the chosen window.

4.1 LAT_CHRONIC_BURDEN

Binary output: LAT_CHRONIC_BURDEN

Label 1 means: high pre-ICU vulnerability or chronic disease burden.

Input variables: age, ICD-9 diagnosis groups, admission type, ICU type, prior admission count if derived before index ICU stay.

Domain indicators

Indicator	Rule
Age burden	age ≥75, or deidentified age category >89
Comorbidity burden	Charlson/Elixhauser count ≥2, or equivalent ICD-derived comorbidity score high
Chronic organ disease	ICD evidence of chronic heart, lung, kidney, liver, neurologic disease
Cancer/immunosuppression	ICD evidence of metastatic cancer, hematologic malignancy, transplant, immunosuppression
High-risk admission context	nonelective admission plus medical ICU or emergency admission source

Score formula: score = sum(indicators)

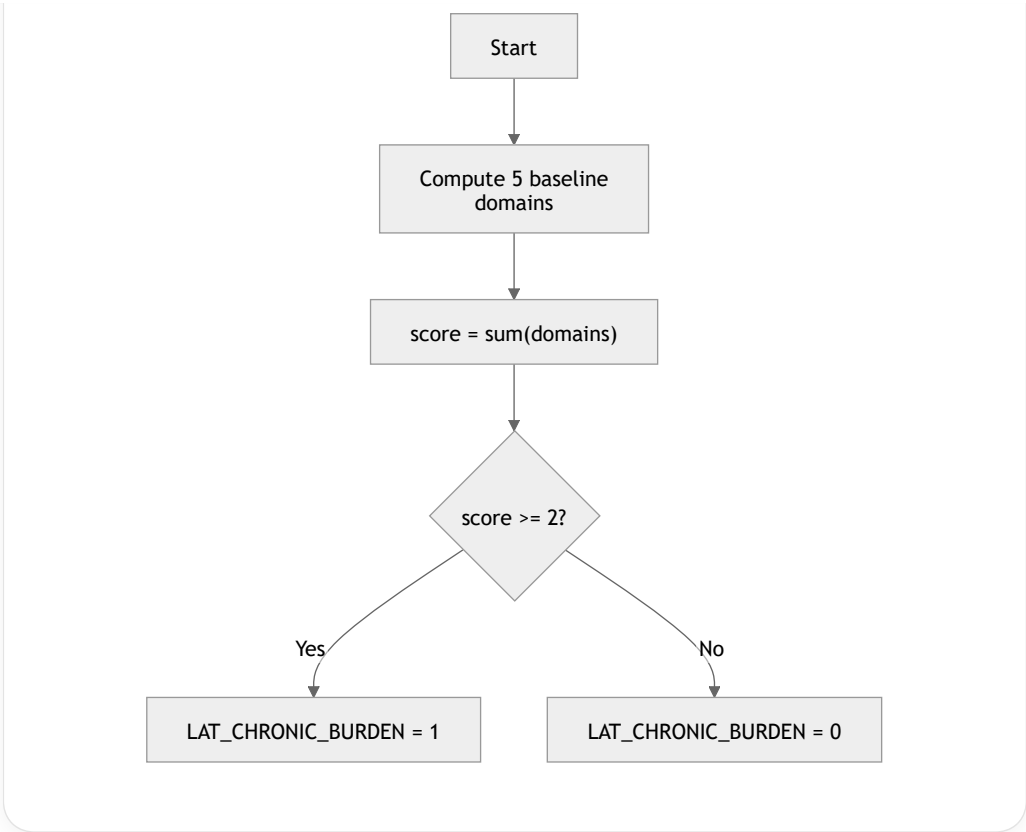
Threshold: LAT_CHRONIC_BURDEN = 1 if score >= 2 else 0

Missingness policy: missing ICD codes are treated as “not documented,” not proof of absence. Add `MISS_ICD_COMORBIDITY_UNKNOWN = 1` if diagnosis coding is unavailable.

</> Python

Run

```
def tag_chronic_burden(x):
    score = 0
    score += int(x.age >= 75 or x.age_is_deidentified_old)
    score += int(x.comorbidity_count >= 2 or x.elixhauser_score >= 5)
    score += int(x.has_chronic_organ_disease)
    score += int(x.has_malignancy_or_immunosuppression)
    score += int(x.admission_type in ["EMERGENCY", "URGENT"] and x.first_careunit in ["MICU", "SICU", "CSRU"])
    return int(score >= 2)
```



Clinical rationale: chronic vulnerability modifies the effect of acute insults and mortality risk.

False positives: age alone may overstate frailty.

False negatives: undercoded comorbidities.

Confidence: high.

4.2 LAT_INFLAMMATION_SEPSIS

Binary output: LAT_INFLAMMATION_SEPSIS

Label 1 means: evidence of systemic inflammation, suspected infection, or sepsis burden.

Input variables: temperature, WBC, lactate, microbiology cultures, antibiotics, admission diagnosis/ICD if within baseline definition.

Domain indicators

Indicator	Rule
Temperature dysregulation	temp_max ≥38.3°C or temp_min <36°C
WBC abnormality	WBC_max > 12 K/μL or WBC_min <4 K/μL
Hypoperfusion/inflammatory stress	lactate_max > 2 mmol/L
Infection testing evidence	blood/urine/respiratory culture ordered or positive culture
Anti-infective treatment evidence	systemic antibiotic prescription started in early window

Score formula: score = sum(indicators)

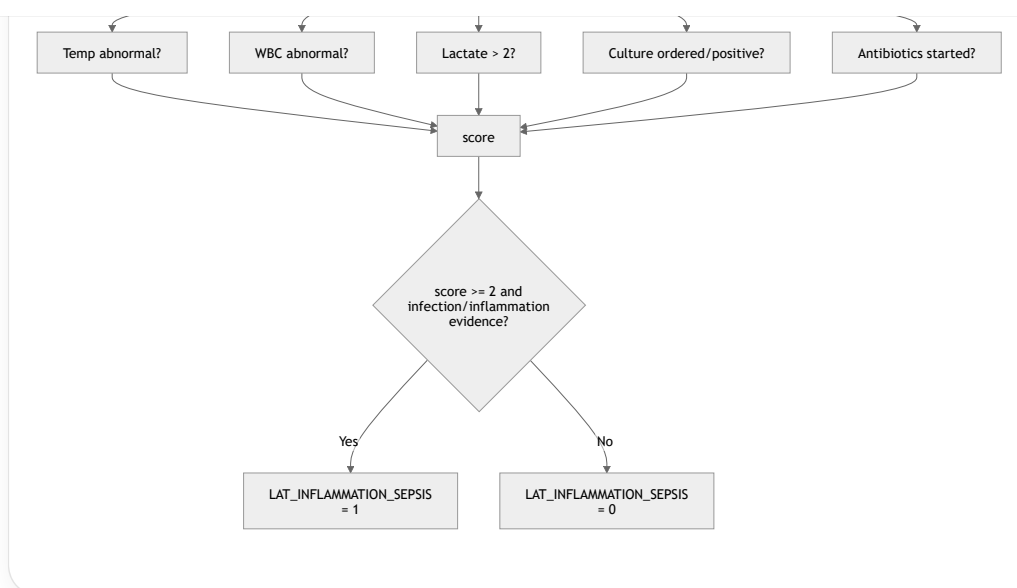
Threshold: LAT_INFLAMMATION_SEPSIS = 1 if score >= 2 and (culture_or_antibiotic or temp_wbc_abnormal) else 0

<> Python

Run

```
def tag_inflammation_sepsis(x):
    temp = x.temp_max_c >= 38.3 or x.temp_min_c < 36
    wbc = x.wbc_max > 12 or x.wbc_min < 4
    lactate = x.lactate_max > 2
    culture = x.culture_ordered or x.culture_positive
    antibiotics = x.systemic_antibiotic_started
    score = sum(map(int, [temp, wbc, lactate, culture, antibiotics]))
    return int(score >= 2 and (culture or antibiotics or temp or wbc))
```


mimic-prompt-running



Clinical rationale: infection and inflammation generate fever, leukocyte abnormalities, lactate elevation, clinician testing, and antibiotics.

False positives: postoperative inflammation, trauma, pancreatitis, prophylactic antibiotics.

False negatives: culture-negative sepsis, immunosuppressed patients without fever/WBC response.

Confidence: medium-high.

4.3 LAT_GLOBAL_SEVERITY

Binary output: LAT_GLOBAL_SEVERITY

Label 1 means: multi-domain acute physiologic severity.

Input variables: MAP/SBP, HR, RR, SpO2, PaO2/FiO2, ventilation, GCS, creatinine, urine output, bilirubin, platelets, lactate, pH, bicarbonate, temperature.

Domain indicators

Indicator	Rule
Circulatory domain	MAP_min < 70 or SBP_min ≤ 100 or vasopressor present
Respiratory domain	SpO2_min < 92, PaO2/FiO2 ≤ 300, or mechanical ventilation
Neurologic domain	GCS_min < 15 or RASS ≤ -3 / ≥ 2
Renal domain	creatinine_max ≥ 2 or urine_output_24h < 500 mL
Metabolic domain	lactate_max > 2, pH_min < 7.30, or bicarbonate_min < 18
Inflammatory domain	temp abnormal or WBC abnormal
Hepatic/coag domain	bilirubin ≥ 2, platelets < 100, or INR ≥ 1.5

Score formula: score = sum(domains)

Threshold: LAT_GLOBAL_SEVERITY = 1 if score ≥ 3 else 0

Python

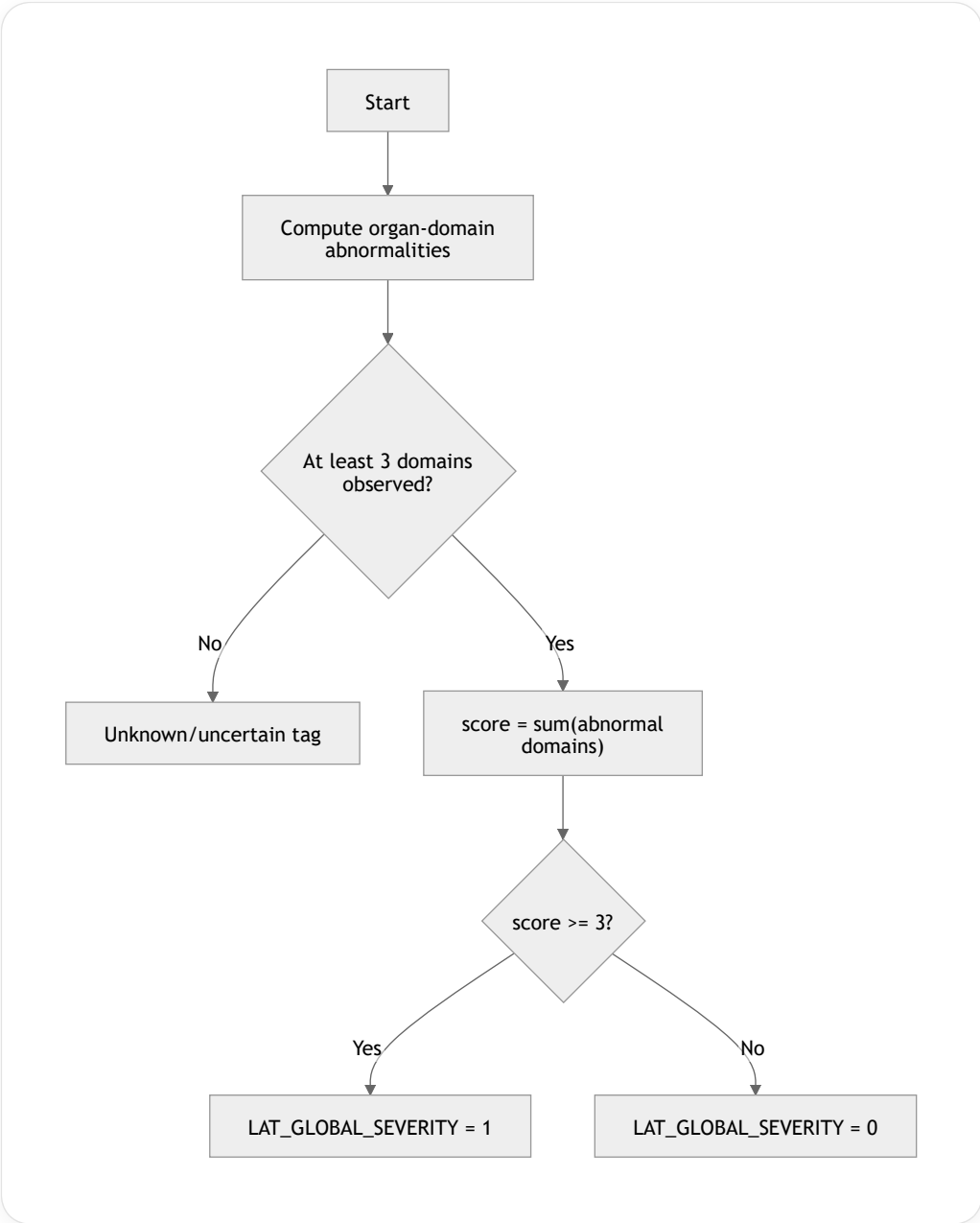


Run

```

def tag_global_severity(x):
    domains = [
        x.map_min < 70 or x.sbp_min <= 100 or x.vasopressor_present,
        x.spo2_min < 92 or x.pao2_fio2_min <= 300 or x.mechanical_ventilation,
        x.gcs_min < 15 or x.rass_min <= -3 or x.rass_max >= 2,
        x.creatinine_max >= 2 or x.urine_output_24h < 500,
        x.lactate_max > 2 or x.ph_min < 7.30 or x.bicarbonate_min < 18,
        x.temp_max_c >= 38.3 or x.temp_min_c < 36 or x.wbc_max > 12 or x.wbc_min < 4,
    ]
  
```

```
x.bilirubin_max >= 2 or x.platelets_min < 100 or x.inr_max >= 1.5,  
]  
observed_domains = [d for d in domains if d is not None]  
if len(observed_domains) < 3:  
    return None  
return int(sum(map(int, observed_domains)) >= 3)
```



Clinical rationale: operationally resembles SOFA/OASIS-style multi-organ severity without using mortality. SOFA explicitly uses respiratory, coagulation, liver, cardiovascular, neurologic, and renal domains. MSD Manuals
False positives: transient abnormalities or treatment-induced values.
False negatives: sparse labs or early unmeasured organ failure.
Confidence: high.

4.4 LAT_SHOCK

Binary output: LAT_SHOCK

Label 1 means: evidence of circulatory shock or clinically meaningful hypoperfusion.

Input variables: MAP, SBP, HR, lactate, vasopressors, urine output, pH/base excess.

Domain indicators

Indicator	Rule
Hypotension	MAP_min <65, MAP_min <70 sustained, or SBP_min ≤90

Indicator	Rule
Vasopressor support	norepinephrine, epinephrine, dopamine, phenylephrine, vasopressin, or dobutamine infusion
Tissue hypoperfusion	$\text{lactate_max} > 2 \text{ mmol/L}$
Renal perfusion consequence	$\text{urine_output_6h} < 0.5 \text{ mL/kg/h}$ or $\text{urine_output_24h} < 500 \text{ mL}$
Acid-base consequence	$\text{pH_min} < 7.30$ or $\text{base_excess_min} \leq -5$

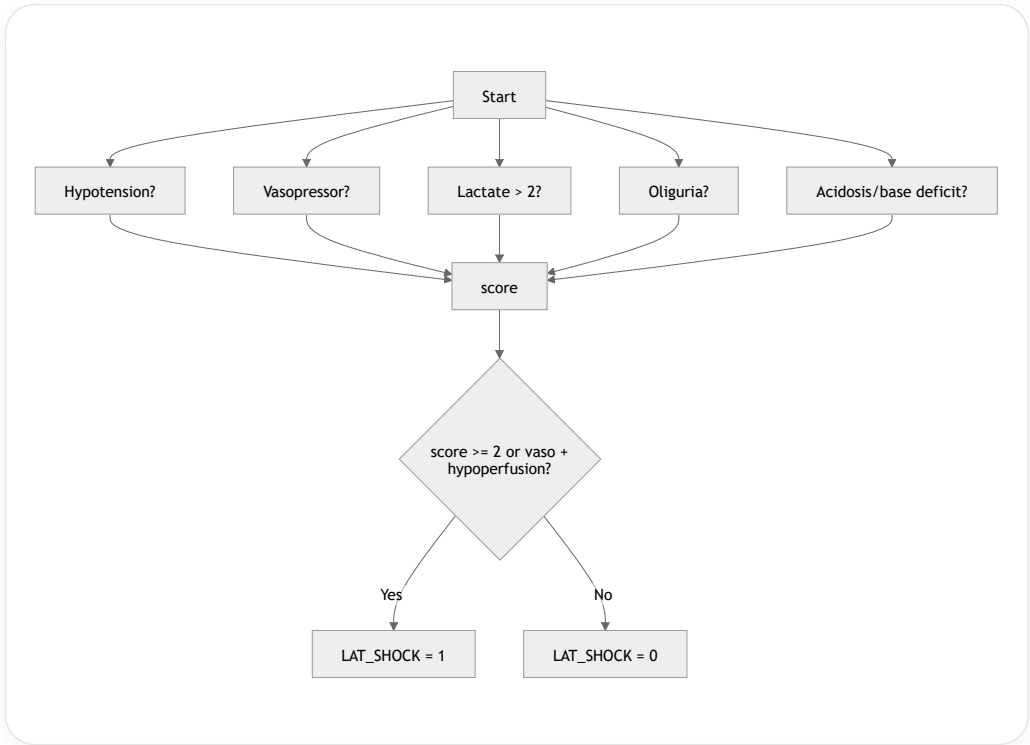
Score formula: $\text{score} = \text{sum}(\text{indicators})$

Threshold: $\text{LAT_SHOCK} = 1$ if $\text{score} \geq 2$ else 0 ; also positive if vasopressor support plus lactate/acidosis.

</> Python

Run

```
def tag_shock(x):  
    hypotension = x.map_min < 65 or x.sbp_min <= 90 or x.map_sustained_lt70  
    vaso = x.vasopressor_present  
    hypoperfusion = x.lactate_max > 2  
    oliguria = x.urine_output_ml_per_kg_hr_6h < 0.5 or x.urine_output_24h < 500  
    acidosis = x.ph_min < 7.30 or x.base_excess_min <= -5  
    score = sum(map(int, [hypotension, vaso, hypoperfusion, oliguria, acidosis]))  
    return int(score >= 2 or (vaso and (hypoperfusion or acidosis or hypotension)))
```



Clinical rationale: Sepsis-3 septic shock uses vasopressor-requiring hypotension and lactate elevation; SOFA cardiovascular scoring uses MAP and vasopressor support. PMC +1

False positives: vasopressors after anesthesia or cardiac surgery.

False negatives: shock treated before ICU arrival or missing lactate.

Confidence: high.

4.5 LAT_RESPIRATORY_FAILURE

Binary output: LAT_RESPIRATORY_FAILURE

Label 1 means: clinically meaningful hypoxemic or ventilatory failure.

Input variables: SpO2, PaO2, FiO2, PaCO2, pH, respiratory rate, mechanical ventilation, PEEP.

Domain indicators

Indicator	Rule
Hypoxemia	$\text{SpO2_min} < 90\text{--}92\%$

Indicator	Rule
Oxygenation failure	PaO2/FiO2_min ≤300
Respiratory support	invasive ventilation, noninvasive ventilation, or FiO2 ≥0.5
Work of breathing	RR_max ≥30 or RR_min ≤8
Ventilatory failure	PaCO2_max ≥50 with pH_min <7.30

Score formula: score = sum(indicators)

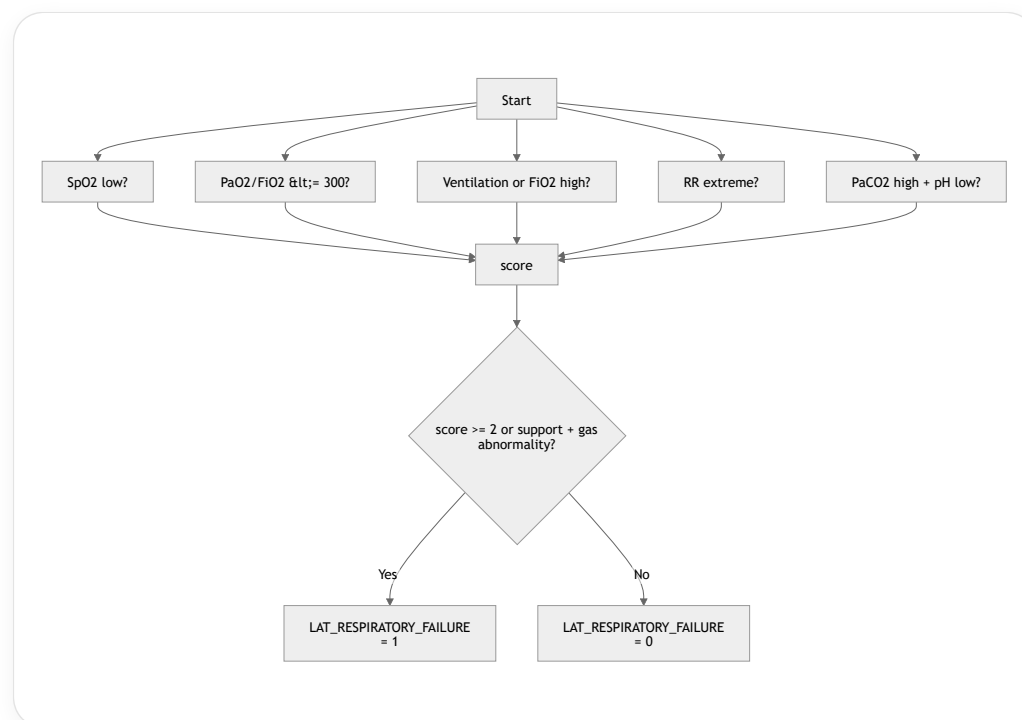
Threshold: LAT_RESPIRATORY_FAILURE = 1 if score ≥ 2 else 0

</> Python



Run

```
def tag_respiratory_failure(x):
    hypoxemia = x.spo2_min < 92
    oxygenation = x.pao2_fio2_min <= 300
    support = x.mechanical_ventilation or x.noninvasive_ventilation or x.fio2_max >= 0.5
    rr_abnormal = x.resp_rate_max >= 30 or x.resp_rate_min <= 8
    hypercapnia = x.paco2_max >= 50 and x.ph_min < 7.30
    score = sum(map(int, [hypoxemia, oxygenation, support, rr_abnormal, hypercapnia]))
    return int(score >= 2 or (support and (oxygenation or hypoxemia or hypercapnia)))
```



Clinical rationale: Berlin ARDS oxygenation categories use PaO2/FiO2 thresholds ≤300, ≤200, and ≤100 with PEEP/CPAP support. MSD Manuals

False positives: oxygen given prophylactically after surgery.

False negatives: missing ABG, pulse oximetry artifact, oxygenation corrected by support.

Confidence: high.

4.6 LAT_RENAL_DYSFUNCTION

Binary output: LAT_RENAL_DYSFUNCTION

Label 1 means: AKI or clinically meaningful renal dysfunction.

Input variables: creatinine, BUN, urine output, potassium, bicarbonate/pH, dialysis procedure.

Domain indicators

Indicator	Rule
Creatinine elevation	creatinine_max ≥2.0 mg/dL, or rise ≥0.3 mg/dL if baseline available
Azotemia	BUN_max ≥40 mg/dL

Indicator	Rule
Oliguria	urine_output_6h <0.5 mL/kg/h or urine_output_24h <500 mL
Renal metabolic consequence	potassium_max ≥5.5 or bicarbonate_min <18
Renal replacement therapy	dialysis/CRRT procedure or dialysis-related chart item

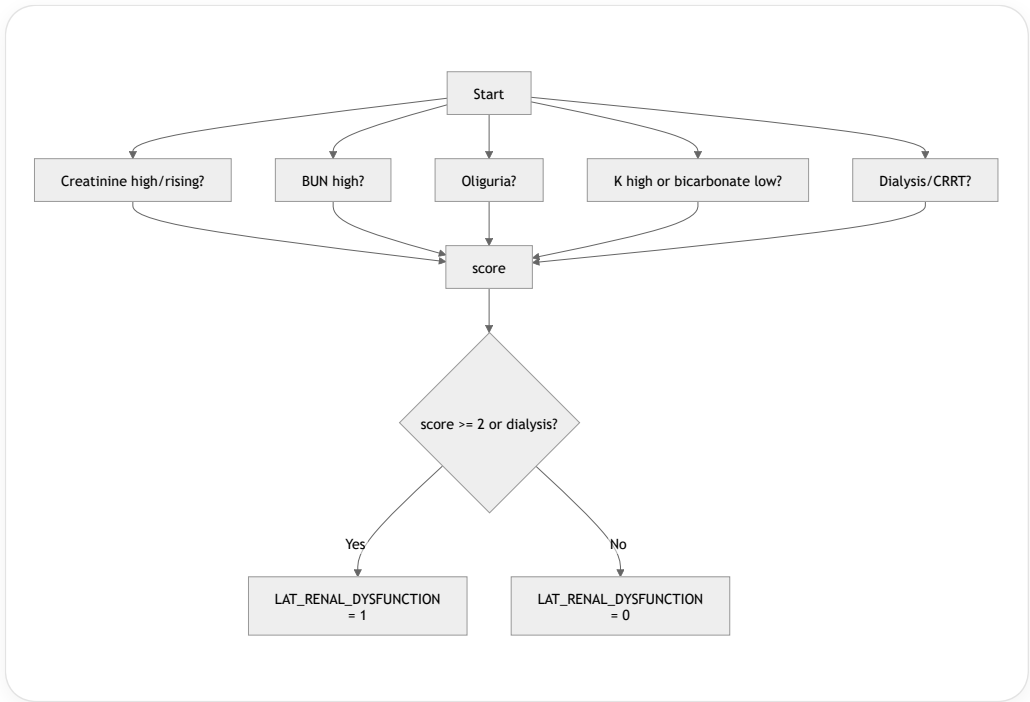
Score formula: score = sum(indicators)

Threshold: LAT_RENAL_DYSFUNCTION = 1 if score >= 2 or dialysis_present else 0

<> Python

Run

```
def tag_renal_dysfunction(x):
    creat = x.creatinine_max >= 2.0 or x.creatinine_delta_from_baseline >= 0.3
    bun = x.bun_max >= 40
    oliguria = x.urine_output_ml_per_kg_hr_6h < 0.5 or x.urine_output_24h < 500
    metabolic = x.potassium_max >= 5.5 or x.bicarbonate_min < 18
    dialysis = x.dialysis_or_crrt_present
    score = sum(map(int, [creat, bun, oliguria, metabolic, dialysis]))
    return int(score >= 2 or dialysis)
```



Clinical rationale: KDIGO AKI staging uses creatinine rise and urine-output decline; severe stages include very high creatinine, oliguria/anuria, and renal replacement therapy. MSD Manuals

False positives: chronic kidney disease without acute injury.

False negatives: no baseline creatinine, uncharted urine output.

Confidence: high.

4.7 LAT_HEPATIC_COAG_DYSFUNCTION

Binary output: LAT_HEPATIC_COAG_DYSFUNCTION

Label 1 means: hepatic, coagulation, or hematologic organ dysfunction.

Input variables: bilirubin, AST, ALT, platelets, INR, PT, PTT, albumin.

Domain indicators

Indicator	Rule
Cholestatic/clearance dysfunction	bilirubin_max ≥2.0 mg/dL
Hepatocellular injury	AST_max or ALT_max ≥3× upper limit of normal, or ≥120 U/L fallback
Thrombocytopenia	platelets_min <100 K/μL

Indicator	Rule
Coagulopathy	INR_max ≥ 1.5 , PT prolonged, or PTT prolonged
Synthetic/chronic liver signal	albumin_min < 2.5 g/dL, only as weak indicator

Score formula: score = sum(indicators)

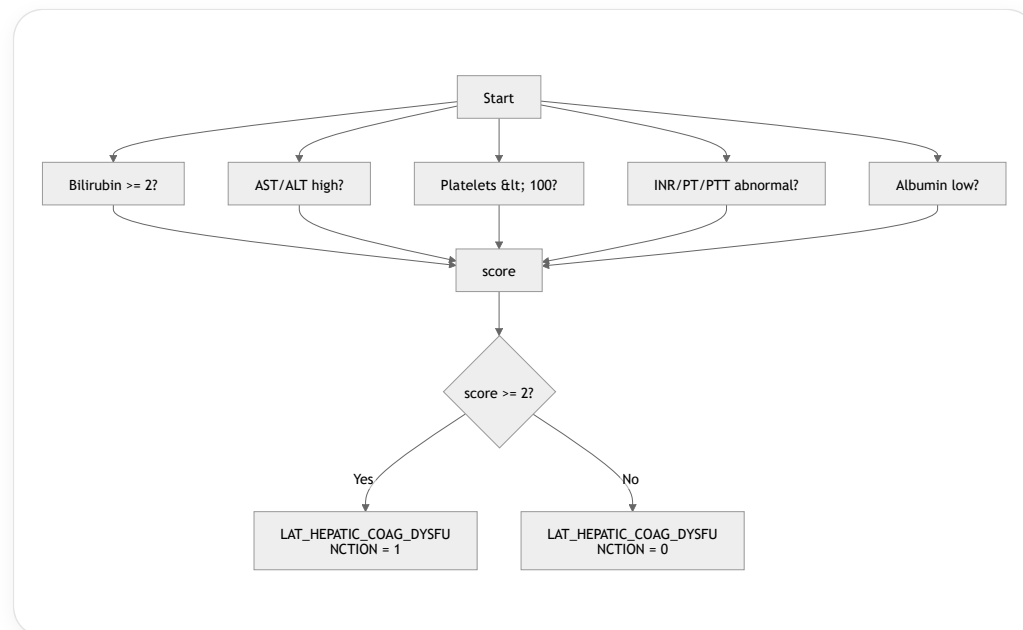
Threshold: LAT_HEPATIC_COAG_DYSFUNCTION = 1 if score ≥ 2 else 0

<> Python



Run

```
def tag_hepatic_coag_dysfunction(x):
    bilirubin = x.bilirubin_max >= 2.0
    transaminitis = x.ast_max >= 120 or x.alt_max >= 120
    platelets = x.platelets_min < 100
    coag = x.inr_max >= 1.5 or x.pt_max_prolonged or x.ptt_max_prolonged
    albumin = x.albumin_min < 2.5
    score = sum(map(int, [bilirubin, transaminitis, platelets, coag, albumin]))
    return int(score >= 2)
```



Clinical rationale: SOFA uses bilirubin for liver dysfunction and platelets for coagulation dysfunction.

MSD Manuals

False positives: anticoagulation, chronic liver disease, dilutional thrombocytopenia.

False negatives: labs not ordered in early window.

Confidence: medium.

4.8 LAT_NEUROLOGIC_DYSFUNCTION

Binary output: LAT_NEUROLOGIC_DYSFUNCTION

Label 1 means: depressed consciousness or acute neurologic dysfunction.

Input variables: GCS total/components, RASS, mental-status charting, pupil response, sedation/intubation indicators.

Domain indicators

Indicator	Rule
GCS depression	GCS_min ≤ 13 ; severe if GCS_min ≤ 8
Motor/verbal/eye abnormality	abnormal GCS component score
RASS abnormality	RASS_min ≤ -3 or RASS_max ≥ 2
Pupillary/neurologic abnormality	abnormal pupil response or focal neurologic charting

Indicator	Rule
Sedation/intubation context	sedative infusion or intubation present; used as confound flag, not positive evidence alone

Score formula: severe GCS counts 2; otherwise score is sum of neuro indicators.

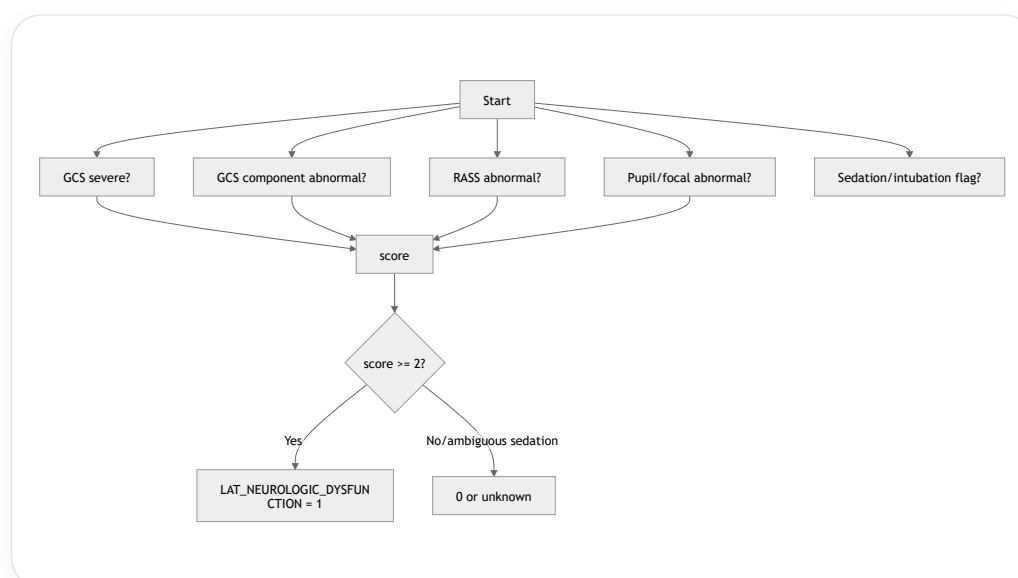
Threshold: LAT_NEUROLOGIC_DYSFUNCTION = 1 if score >= 2 else 0

</> Python



Run

```
def tag_neurologic_dysfunction(x):
    score = 0
    if x.gcs_min <= 8:
        score += 2
    elif x.gcs_min <= 13:
        score += 1
    score += int(x.gcs_component_abnormal)
    score += int(x.rass_min <= -3 or x.rass_max >= 2)
    score += int(x.pupil_or_focal_neuro_abnormal)
    # sedation flag is not positive by itself; it marks ambiguity
    if x.sedation_or_intubation and score == 1:
        return None
    return int(score >= 2)
```



Clinical rationale: neurologic SOFA is based on GCS, but ICU sedation can mask neurologic status.

False positives: sedatives, postoperative anesthesia, intubation limiting verbal response.

False negatives: missing GCS or neuro exams.

Confidence: medium-high.

4.9 LAT_METABOLIC_DERANGEMENT

Binary output: LAT_METABOLIC_DERANGEMENT

Label 1 means: clinically meaningful acid-base, lactate, glucose, or electrolyte instability.

Input variables: pH, bicarbonate, base excess, lactate, sodium, potassium, glucose, anion gap if derived.

Domain indicators

Indicator	Rule
Acidemia/alkalemia	pH_min <7.30 or pH_max >7.55
Bicarbonate/base disturbance	bicarbonate_min <18, bicarbonate_max >35, or base_excess_min ≤-5
Lactate stress	lactate_max > 2 mmol/L
Potassium disorder	potassium_min <3.0 or potassium_max ≥5.5
Sodium disorder	sodium_min <130 or sodium_max >150

Indicator	Rule
Glucose disorder	glucose_min <70 or glucose_max >250 mg/dL

Score formula: score = sum(indicators)

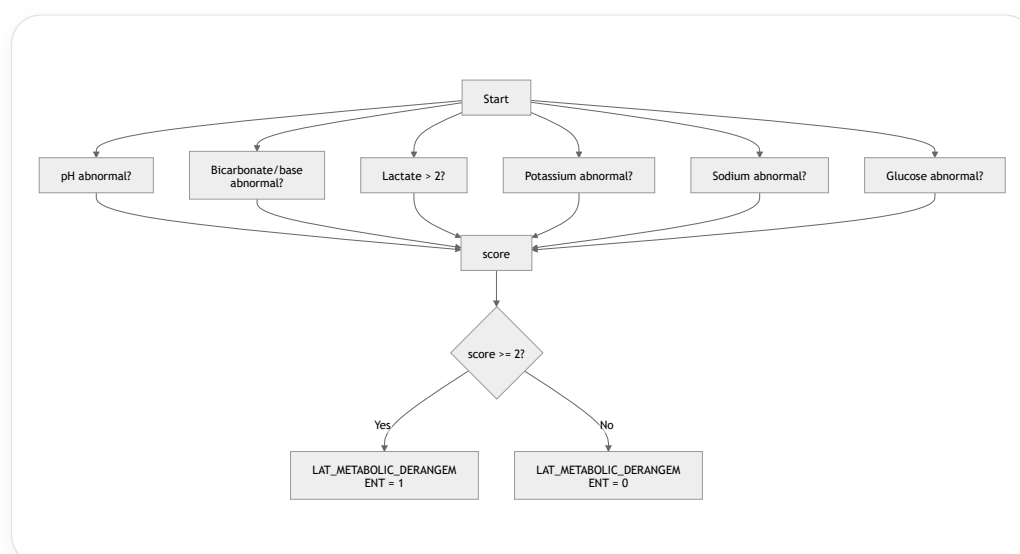
Threshold: LAT_METABOLIC_DERANGEMENT = 1 if score >= 2 else 0

</> Python



Run

```
def tag_metabolic_derangement(x):
    ph = x.ph_min < 7.30 or x.ph_max > 7.55
    bicarb = x.bicarbonate_min < 18 or x.bicarbonate_max > 35 or x.base_excess_min <= -5
    lactate = x.lactate_max > 2
    potassium = x.potassium_min < 3.0 or x.potassium_max >= 5.5
    sodium = x.sodium_min < 130 or x.sodium_max > 150
    glucose = x.glucose_min < 70 or x.glucose_max > 250
    score = sum(map(int, [ph, bicarb, lactate, potassium, sodium, glucose]))
    return int(score >= 2)
```



Clinical rationale: metabolic derangement is a downstream mediator of shock, renal dysfunction, respiratory failure, and sepsis.

False positives: chronic renal failure, diabetic ketoacidosis, treatment-induced electrolyte shifts.

False negatives: missing ABG or chemistry panels.

Confidence: high.

4.10 LAT_CARDIAC_STRAIN

Binary output: LAT_CARDIAC_STRAIN

Label 1 means: evidence of myocardial injury, strain, or major rhythm stress.

Input variables: troponin, CK-MB, heart rate, arrhythmia charting, ECG report metadata, vasopressor/cardiac ICU context as weak contextual signal.

Domain indicators

Indicator	Rule
Myocardial biomarker	troponin above local assay upper reference limit, or troponin qualitatively abnormal
CK-MB injury signal	CK-MB above reference range
Rhythm instability	arrhythmia charting, atrial/ventricular tachyarrhythmia, severe bradycardia
Hemodynamic cardiac stress	HR_max > 130 or HR_min < 40 with hypotension
Cardiac-care context	CSRU/CCU/cardiac surgery context, weak indicator only

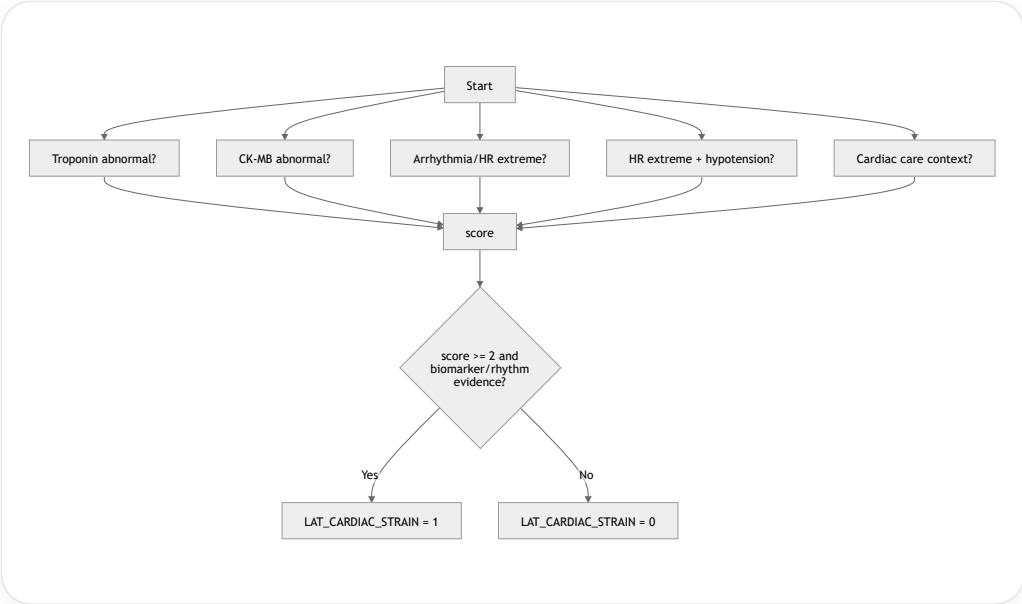
Score formula: biomarker or rhythm evidence required; score = sum(indicators)

Threshold: LAT_CARDIAC_STRAIN = 1 if score >= 2 and (biomarker or rhythm) else 0

</> Python

Run

```
def tag_cardiac_strain(x):
    biomarker = x.troponin_above_u1n or x.troponin_flag_abnormal
    ckmb = x.ckmb_above_u1n
    rhythm = x.arrhythmia_documented or x.hr_max > 150 or x.hr_min < 40
    hemodynamic = (x.hr_max > 130 or x.hr_min < 40) and (x.map_min < 70 or x.sbp_min <= 100)
    context = x.first_careunit in ["CSRU", "CCU"] or x.cardiac_surgery_context
    score = sum(map(int, [biomarker, ckmb, rhythm, hemodynamic, context]))
    return int(score >= 2 and (biomarker or rhythm))
```



Clinical rationale: cardiac injury can precipitate shock and global severity, but measurement is selective.
False positives: demand ischemia, renal failure elevating troponin, postoperative cardiac surgery.
False negatives: troponin not ordered.
Confidence: low-medium.

5. Observed-to-latent measurement model

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
	LAT_CHRONIC_BURDEN / background	Older → higher vulnerability	Nonspecific	Yes	No	De dei spe
/gender	Background	Context-dependent	Nonspecific	No	Possible	Co not
nicity/insurance/langua	Background/social context	Context-dependent	Nonspecific	No	Yes	Ma me
mission type/location	Background	Emergency/urgent → higher acute risk	Nonspecific	Yes	Yes	Co ind
type	Background/care context	Unit-specific risk	Nonspecific	Yes	Yes	Car me
-9 diagnoses	LAT_CHRONIC_BURDEN , admission phenotype	Chronic disease codes increase burden	Medium	Yes	Coding process	No out
irt rate	LAT_SHOCK , LAT_INFLAMMATION_SEPSIS , LAT_CARDIAC_STRAIN , LAT_GLOBAL_SEVERITY	High or very low abnormal	Nonspecific	Yes	Yes	Pai shc

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
P/SBP	LAT_SHOCK , LAT_GLOBAL_SEVERITY	Low abnormal; very high may severity	Medium	Yes	Strongly treatment-modified	Vas cha
piratory rate	LAT_RESPIRATORY_FAILURE , LAT_METABOLIC_DERANGEMENT , LAT_INFLAMMATION_SEPSIS	High/low abnormal	Nonspecific	Yes	Yes	Co anc
2	LAT_RESPIRATORY_FAILURE	Low abnormal	Medium	Yes	Oxygen therapy modifies	Ser
2/FiO2	LAT_RESPIRATORY_FAILURE	Lower worse	Specific	Yes	Ventilator/FiO2 dependent	Rec FiC
2/pH	LAT_RESPIRATORY_FAILURE , LAT_METABOLIC_DERANGEMENT	Hypercapnia with low pH worse	Medium	Yes	ABG ordering informative	Stri
perature	LAT_INFLAMMATION_SEPSIS , LAT_GLOBAL_SEVERITY	Fever/hypothermia abnormal	Nonspecific	Yes	Frequent vitals	Hyl ICL
C	LAT_INFLAMMATION_SEPSIS	High/low abnormal	Nonspecific	Yes	Lab ordering	Ste cor
tate	LAT_SHOCK , LAT_INFLAMMATION_SEPSIS , LAT_METABOLIC_DERANGEMENT , LAT_GLOBAL_SEVERITY	Higher worse	Medium	Yes	Highly order-dependent	Stri ma
atinine	LAT_RENAL_DYSFUNCTION	Higher/rising worse	Medium	Yes	Lab ordering	Chi cor
4	LAT_RENAL_DYSFUNCTION , LAT_CHRONIC_BURDEN	Higher worse	Nonspecific	Yes	Lab ordering	Als ble
re output	LAT_RENAL_DYSFUNCTION , LAT_SHOCK	Lower worse	Medium	Yes	Charting and catheter use	Ou inf
assium	LAT_RENAL_DYSFUNCTION , LAT_METABOLIC_DERANGEMENT	High/low abnormal	Medium	Yes	Treatment-modified	Dia affe
arbonate/base excess	LAT_METABOLIC_DERANGEMENT , LAT_RENAL_DYSFUNCTION , LAT_SHOCK	Low or extreme abnormal	Medium	Yes	Lab/ABG ordering	Acid
lium	LAT_METABOLIC_DERANGEMENT	Low/high abnormal	Nonspecific	Yes	Treatment/fluid modified	Flu
cose	LAT_METABOLIC_DERANGEMENT , LAT_INFLAMMATION_SEPSIS	Low/high abnormal	Nonspecific	Yes	Frequent if diabetic/insulin	Me inf
elelets	LAT_HEPATIC_COAG_DYSFUNCTION , LAT_INFLAMMATION_SEPSIS	Lower worse	Medium	Yes	Lab ordering	Seq dilt
/PT/PTT	LAT_HEPATIC_COAG_DYSFUNCTION	Higher/prolonged worse	Medium	Yes	Anticoagulants/procedures	Stri cor
ubin	LAT_HEPATIC_COAG_DYSFUNCTION	Higher worse	Medium	Yes	Selectively ordered	SO
/ALT	LAT_HEPATIC_COAG_DYSFUNCTION , LAT_SHOCK	Higher worse	Nonspecific	Yes	Selectively ordered	Sho
umin	LAT_CHRONIC_BURDEN , LAT_HEPATIC_COAG_DYSFUNCTION	Lower worse	Nonspecific	Weak	Selectively ordered	Chi infl ma
5	LAT_NEUROLOGIC_DYSFUNCTION , LAT_GLOBAL_SEVERITY	Lower worse	Medium	Yes	Sedation/intubation	Ne

served variable	Parent latent variable(s)	Direction of abnormality	Specificity	Used in tree?	Care-process / measurement-intensity influence?	No
is	LAT_NEUROLOGIC_DYSFUNCTION , care process	Extreme low/high abnormal	Medium	Yes	Sedation target	No
ionin/CK-MB	LAT_CARDIAC_STRAIN , LAT_RENAL_DYSFUNCTION	Higher worse	Medium	Yes	Selectively ordered	Spa cor
opressor administration	TRT_VASOPRESSORS , proxy for LAT_SHOCK	Present → shock severity proxy	Specific for treatment, not biology	Yes	Assignment by clinician	Tre me
chanical ventilation	TRT_MECH_VENT , proxy for LAT_RESPIRATORY_FAILURE	Present → severe respiratory/airway support	Medium	Yes	Clinical decision	Car tha
lysis/CRRT	TRT_DIALYSIS , proxy for LAT_RENAL_DYSFUNCTION	Present → severe renal/metabolic failure	High	Yes	Clinical decision	Tre ma
ibiotics/cultures	TRT_ANTIBIOTICS , proxy for LAT_INFLAMMATION_SEPSIS	Present/ordered → suspected infection	Medium	Yes	Clinician suspicion	No inf
/vital counts	MISS_MEASUREMENT_INTENSITY	More frequent often higher severity	Nonspecific	Limited	Directly	Inf bio

6. Latent causal DAG edge list

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
BG_AGE	LAT_CHRONIC_BURDEN	background-to-latent	Aging increases chronic vulnerability and frailty burden	High	Clinical plausibility
BG_SEX	LAT_CHRONIC_BURDEN	background-to-latent	Sex differences in comorbidity and physiology	Medium	Case-mix
BG_ETHNICITY_INSURANCE_L ANGUAGE	MISS_MEASUREMENT_INTENS ITY	measurement-process	Social/access factors may affect care and documentation	Medium	Informative observation literature
BG_ADMISSION_CONTEXT	LAT_INFLAMMATION_SEPSI S	background-to-latent	Emergency/medical admissions more likely acute infection	Medium	Case-mix
BG_ADMISSION_CONTEXT	LAT_GLOBAL_SEVERITY	background-to-latent	Admission source/type captures acute presentation	Medium	Case-mix
BG_ICU_UNIT	LAT_GLOBAL_SEVERITY	background-to-latent	ICU type reflects illness mix	Medium	Dataset support
BG_ICU_UNIT	MISS_MEASUREMENT_INTENS ITY	measurement-process	Units differ in monitoring and charting	High	Dataset source/care process
LAT_CHRONIC_BURDEN	LAT_GLOBAL_SEVERITY	latent-to-latent	Vulnerability worsens physiologic response to acute illness	High	Clinical mechanism
LAT_CHRONIC_BURDEN	OUT_MORTALITY	latent-to-mortality	Frailty/comorbidity directly affects survival reserve	High	Clinical mechanism
LAT_INFLAMMATION_SEPSIS	LAT_GLOBAL_SEVERITY	latent-to-latent	Systemic inflammation drives multi-organ instability	High	Sepsis/organ dysfunction
LAT_INFLAMMATION_SEPSIS	LAT_SHOCK	latent-to-latent	Vasodilation, capillary leak, myocardial depression	High	Septic shock mechanism

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
LAT_INFLAMMATION_SEPSIS	LAT_HEPATIC_COAG_DYSFUNCTION	latent-to-latent	Sepsis-associated thrombocytopenia/coagulopathy/liver injury	Medium	Clinical mechanism
LAT_GLOBAL_SEVERITY	LAT_RESPIRATORY_FAILURE	latent-to-latent	Severe illness can lead to hypoxemia/ventilatory failure	Medium	ICU physiology
LAT_GLOBAL_SEVERITY	LAT_NEUROLOGIC_DYSFUNCTION	latent-to-latent	Encephalopathy, sedation need, hypoperfusion	Medium	ICU physiology
LAT_GLOBAL_SEVERITY	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Multi-organ severity disrupts homeostasis	High	Clinical mechanism
LAT_CARDIAC_STRAIN	LAT_SHOCK	latent-to-latent	Pump failure or arrhythmia can cause hypoperfusion	Medium	Clinical mechanism
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	latent-to-latent	Renal hypoperfusion and ischemic tubular injury	High	Clinical mechanism
LAT_SHOCK	LAT_HEPATIC_COAG_DYSFUNCTION	latent-to-latent	Shock liver, coagulopathy, DIC risk	Medium	Clinical mechanism
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Lactate/acidosis from hypoperfusion	High	Clinical mechanism
LAT_RESPIRATORY_FAILURE	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hypercapnia/acidosis and hypoxemia	High	Clinical mechanism
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	latent-to-latent	Hyperkalemia, acidosis, uremia	High	KDIGO/metabo mechanism
LAT_SHOCK	OUT_MORTALITY	latent-to-mortality	Inadequate perfusion causes organ failure/death	High	Clinical mechanism
LAT_RESPIRATORY_FAILURE	OUT_MORTALITY	latent-to-mortality	Hypoxemia/hypercapnia and ventilatory failure	High	Clinical mechanism
LAT_RENAL_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	AKI reflects and worsens multi-organ failure	High	Clinical mechanism
LAT_HEPATIC_COAG_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	Bleeding, thrombosis, liver failure, severe systemic illness	Medium	SOFA organ domains
LAT_NEUROLOGIC_DYSFUNCTION	OUT_MORTALITY	latent-to-mortality	Coma/encephalopathy/brain injury indicates reduced survival	High	SOFA/OASIS domains
LAT_METABOLIC_DERANGEMENT	OUT_MORTALITY	latent-to-mortality	Severe acidosis/electrolyte/glucose derangement can be fatal	High	Clinical mechanism
LAT_CARDIAC_STRAIN	OUT_MORTALITY	latent-to-mortality	Myocardial injury/arrhythmia increases mortality	Medium	Clinical mechanism
LAT_*	OBS_*	latent-to-observed	Latent states generate labs/vitals/charted values	High	Measurement model
LAT_GLOBAL_SEVERITY	MISS_MEASUREMENT_INTENSITY	measurement-process	Sicker patients receive more tests and monitoring	High	Informative observation literature
LAT_INFLAMMATION_SEPSIS	TRT_ANTIBIOTICS	treatment/care-process	Suspected infection causes antibiotic ordering	High	Clinical care process
LAT_SHOCK	TRT_VASOPRESSORS	treatment/care-process	Shock causes vasopressor treatment	High	Clinical care process

Source node	Target node	Edge type	Clinical mechanism	Confidence	Reasoning basi
LAT_RESPIRATORY_FAILURE	TRT_MECH_VENT	treatment/care-process	Respiratory failure causes ventilatory support	High	Clinical care process
LAT_RENAL_DYSFUNCTION	TRT_DIALYSIS	treatment/care-process	Severe renal/metabolic failure causes dialysis	High	Clinical care process
TRT_*	OBS_*	treatment-to-observed	Treatments change observed vitals/labs	Medium	Treatment response
MISS_MEASUREMENT_INTENSITY	OBS_AVAILABILITY	measurement-process	Ordering/charting determines whether measurements exist	High	EHR process

7. Mortality mechanism summary

Mortality in this DAG is caused by **latent clinical states**, not raw measurements. The main mortality pathways are:

1. Chronic vulnerability pathway:
BG_AGE / BG_ADMISSION_CONTEXT → LAT_CHRONIC_BURDEN → LAT_GLOBAL_SEVERITY → organ dysfunction → OUT_MORTALITY .
2. Sepsis/shock pathway:
LAT_INFLAMMATION_SEPSIS → LAT_SHOCK → LAT_RENAL_DYSFUNCTION / LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY .
3. Respiratory failure pathway:
LAT_RESPIRATORY_FAILURE → hypoxemia/hypercapnia/acidosis represented by LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY .
4. Renal-metabolic pathway:
LAT_RENAL_DYSFUNCTION → LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY .
5. Cardiac-hemodynamic pathway:
LAT_CARDIAC_STRAIN → LAT_SHOCK → LAT_GLOBAL_SEVERITY / LAT_METABOLIC_DERANGEMENT → OUT_MORTALITY .

The observed values are children of these states. For example, lactate is an observed proxy of shock/metabolic stress; it is not modeled as the root cause of death. Similarly, low PaO2/FiO2 indicates respiratory failure, and creatinine indicates renal dysfunction.

8. Missingness and irregular sampling analysis

DATASET-NAME is not sampled by a fixed research protocol. It is routine-care EHR data: nurses chart vitals, clinicians order labs, medications are recorded when administered or prescribed, and some procedures are recorded only in specific source systems. The official documentation states that the database was populated from routine hospital care and includes time-stamped bedside measurements, medications, fluid balances, labs, and reports. physionet.org

Missingness is therefore rarely pure MCAR. The most plausible mechanisms are:

Variable class	Likely missingness mechanism	Interpretation
Hourly vitals	MAR/care-intensity-driven	Usually frequent in ICU, but frequency still reflects acuity and unit practice.
ABGs, lactate, troponin, coagulation labs	MNAR / informative observation	Ordered when clinicians suspect hypoxemia, shock, cardiac injury, bleeding, or severe illness.
Microbiology cultures	Clinician-suspicion-driven	Absence of culture does not prove absence of infection.
Urine output	Care-process dependent	Requires charting and often catheterization.
Ventilator settings	Treatment-dependent	Missing may mean not ventilated or simply unrecorded depending extraction logic.

Variable class	Likely missingness mechanism	Interpretation
ICD diagnoses	Coding process	No timestamp; missing code is not equivalent to absence.
Notes/reports	Care-process and documentation-driven	Highly informative but can contain leakage if discharge summaries are used.

Informative presence and informative observation are well-described in routine health data: the presence, timing, and frequency of observations can themselves carry information about health state, and the literature specifically notes that more severe conditions often lead to more intense monitoring. OUP Academic

For latent labels:

- Do **not** treat missing high-cost/selective labs as normal.
- Use missingness indicators and observation counts as separate features.
- Allow “unknown” or low-confidence tags when too few domains are observed.
- Use measurement frequency only as a weak auxiliary signal in `LAT_GLOBAL_SEVERITY` ; do not let missingness dominate biological definitions.
- Represent `MISS_MEASUREMENT_INTENSITY` as a separate node influenced by severity, ICU unit, clinician suspicion, and care processes.

Bias risks:

- Conditioning on measurement availability can create collider bias because severity and care unit both affect whether labs are measured.
- Using discharge summaries or post-outcome notes can create leakage.
- Using treatment presence as if it were pure disease severity can induce confounding by indication.

9. Backdoor/confounding implications

For causal effects on mortality, adjustment depends on the exposure.

Likely confounders

Exposure type	Candidate confounders
Baseline exposure, e.g. admission type or ICU unit	Age, sex, ethnicity/social-context proxies, chronic burden, admission source
Early treatment, e.g. vasopressors	Chronic burden, admission context, initial shock severity, sepsis burden, cardiac strain
Ventilation strategy	Chronic lung disease, admission type, respiratory failure severity, neurologic dysfunction, surgical context
Antibiotics	Infection suspicion, temperature/WBC, cultures, chronic immunosuppression, admission diagnosis
Dialysis	Baseline kidney disease, renal dysfunction severity, shock, metabolic derangement

Likely mediators

Variable	Why mediator
<code>LAT_SHOCK</code>	Mediates sepsis/cardiac injury effects on mortality.
<code>LAT_RENAL_DYSFUNCTION</code>	Mediates shock/sepsis effects on mortality.
<code>LAT_METABOLIC_DERANGEMENT</code>	Mediates shock, renal, and respiratory effects.
<code>LAT_GLOBAL_SEVERITY</code>	Often summarizes downstream acute physiology; inappropriate adjustment for total effects of upstream exposures.
Treatments	May mediate effects of latent severity on later outcomes, while also being confounded by severity.

Colliders and variables to avoid conditioning on

Variable	Collider risk
MISS_MEASUREMENT_INTENSITY	Caused by latent severity, clinician suspicion, ICU unit, and social/care factors.
Lab availability	Conditioning on ordered labs selects on clinician suspicion and severity.
ICU length of stay	Affected by survival, severity, and discharge readiness.
Discharge location	Downstream of outcome trajectory.
LAT_GLOBAL_SEVERITY	Can be a collider/proxy outcome if estimating effects of upstream disease states or treatments.
Treatment receipt	Collider/confounder-by-indication if conditioned on without longitudinal causal design.

For estimating the **total effect** of an upstream exposure such as sepsis burden on mortality, do not adjust for downstream `LAT_SHOCK` , `LAT_RENAL_DYSFUNCTION` , `LAT_METABOLIC_DERANGEMENT` , or `LAT_GLOBAL_SEVERITY` . For estimating a **controlled direct effect**, adjustment may be appropriate only with a clearly specified target trial and temporal ordering.

For ICU time-varying treatments, standard regression adjustment is often biased when prior treatment affects later confounders; ICU causal inference reviews emphasize target-trial framing and G-methods for treatment-affected time-varying confounding. Nature

Binary latent tags are therefore not automatically safe adjustment variables. They may be:

- confounders when measured before exposure,
- mediators when downstream of exposure,
- colliders when defined using measurement intensity or treatment decisions,
- proxy outcomes when they are outcome-proximal severity markers.

10. Graph specification

A. Adjacency list

BG_AGE:

- LAT_CHRONIC_BURDEN
- OUT_MORTALITY

BG_SEX:

- LAT_CHRONIC_BURDEN

BG_ETHNICITY_INSURANCE_LANGUAGE:

- MISS_MEASUREMENT_INTENSITY
- OUT_MORTALITY

BG_ADMISSION_CONTEXT:

- LAT_INFLAMMATION_SEPSIS
- LAT_GLOBAL_SEVERITY
- MISS_MEASUREMENT_INTENSITY

BG_ICU_UNIT:

- LAT_GLOBAL_SEVERITY
- MISS_MEASUREMENT_INTENSITY

LAT_CHRONIC_BURDEN:

- LAT_GLOBAL_SEVERITY
- LAT_RENAL_DYSFUNCTION
- LAT_CARDIAC_STRAIN
- OUT_MORTALITY

LAT_INFLAMMATION_SEPSIS:

- LAT_GLOBAL_SEVERITY
- LAT_SHOCK
- LAT_HEPATIC_COAG_DYSFUNCTION
- TRT_ANTIBIOTICS
- OUT_MORTALITY

LAT_GLOBAL_SEVERITY:

- LAT_RESPIRATORY_FAILURE
- LAT_NEUROLOGIC_DYSFUNCTION
- LAT_METABOLIC_DERANGEMENT
- MISS_MEASUREMENT_INTENSITY
- OUT_MORTALITY

LAT_CARDIAC_STRAIN:

- LAT_SHOCK
- OUT_MORTALITY

LAT_SHOCK:

- LAT_RENAL_DYSFUNCTION
- LAT_HEPATIC_COAG_DYSFUNCTION
- LAT_METABOLIC_DERANGEMENT
- TRT_VASOPRESSORS
- OUT_MORTALITY

LAT_RESPIRATORY_FAILURE:

- LAT_METABOLIC_DERANGEMENT
- TRT_MECH_VENT
- OUT_MORTALITY

LAT_RENAL_DYSFUNCTION:

- LAT_METABOLIC_DERANGEMENT
- TRT_DIALYSIS
- OUT_MORTALITY

LAT_HEPATIC_COAG_DYSFUNCTION:

- OUT_MORTALITY

LAT_NEUROLOGIC_DYSFUNCTION:

- OUT_MORTALITY

LAT_METABOLIC_DERANGEMENT:

- OUT_MORTALITY

TRT_ANTIBIOTICS:

- OBS_CULTURES_MEDICATIONS

TRT_VASOPRESSORS:

- OBS_BLOOD_PRESSURE
- OBS_LACTATE
- OUT_MORTALITY

TRT_MECH_VENT:

- OBS_OXYGENATION
- OBS_VENTILATOR_SETTINGS
- OUT_MORTALITY

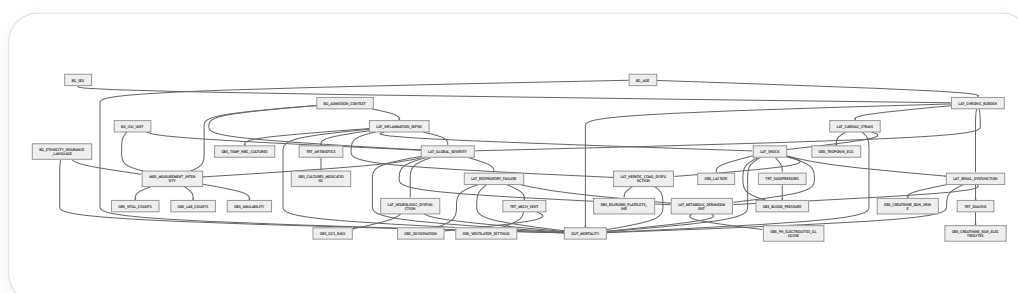
TRT_DIALYSIS:

- OBS_CREATININE_BUN_ELECTROLYTES
- OUT_MORTALITY

MISS_MEASUREMENT_INTENSITY:

- OBS_AVAILABILITY
- OBS_LAB_COUNTS
- OBS_VITAL_COUNTS

B. Mermaid graph



C. Python NetworkX edge list

</> Python



Run

```

edges = [
    ("BG_AGE", "LAT_CHRONIC_BURDEN"),
    ("BG_AGE", "OUT_MORTALITY"),
    ("BG_SEX", "LAT_CHRONIC_BURDEN"),
    ("BG_ETHNICITY_INSURANCE_LANGUAGE", "MISS_MEASUREMENT_INTENSITY"),
    ("BG_ETHNICITY_INSURANCE_LANGUAGE", "OUT_MORTALITY"),
    ("BG_ADMISSION_CONTEXT", "LAT_INFLAMMATION_SEPSIS"),
    ("BG_ADMISSION_CONTEXT", "LAT_GLOBAL_SEVERITY"),
    ("BG_ADMISSION_CONTEXT", "MISS_MEASUREMENT_INTENSITY"),
    ("BG_ICU_UNIT", "LAT_GLOBAL_SEVERITY"),
    ("BG_ICU_UNIT", "MISS_MEASUREMENT_INTENSITY"),

    ("LAT_CHRONIC_BURDEN", "LAT_GLOBAL_SEVERITY"),
    ("LAT_CHRONIC_BURDEN", "LAT_RENAL_DYSFUNCTION"),
    ("LAT_CHRONIC_BURDEN", "LAT_CARDIAC_STRAIN"),
    ("LAT_CHRONIC_BURDEN", "OUT_MORTALITY"),

    ("LAT_INFLAMMATION_SEPSIS", "LAT_GLOBAL_SEVERITY"),
    ("LAT_INFLAMMATION_SEPSIS", "LAT_SHOCK"),
    ("LAT_INFLAMMATION_SEPSIS", "LAT_HEPATIC_COAG_DYSFUNCTION"),
    ("LAT_INFLAMMATION_SEPSIS", "TRT_ANTIBIOTICS"),
    ("LAT_INFLAMMATION_SEPSIS", "OUT_MORTALITY"),

    ("LAT_GLOBAL_SEVERITY", "LAT_RESPIRATORY_FAILURE"),
    ("LAT_GLOBAL_SEVERITY", "LAT_NEUROLOGIC_DYSFUNCTION"),
    ("LAT_GLOBAL_SEVERITY", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_GLOBAL_SEVERITY", "MISS_MEASUREMENT_INTENSITY"),
    ("LAT_GLOBAL_SEVERITY", "OUT_MORTALITY"),

    ("LAT_CARDIAC_STRAIN", "LAT_SHOCK"),
    ("LAT_CARDIAC_STRAIN", "OUT_MORTALITY"),

    ("LAT_SHOCK", "LAT_RENAL_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_HEPATIC_COAG_DYSFUNCTION"),
    ("LAT_SHOCK", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_SHOCK", "TRT_VASOPRESSORS"),
    ("LAT_SHOCK", "OUT_MORTALITY"),

    ("LAT_RESPIRATORY_FAILURE", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RESPIRATORY_FAILURE", "TRT_MECH_VENT"),
    ("LAT_RESPIRATORY_FAILURE", "OUT_MORTALITY"),

    ("LAT_RENAL_DYSFUNCTION", "LAT_METABOLIC_DERANGEMENT"),
    ("LAT_RENAL_DYSFUNCTION", "TRT_DIALYSIS"),
    ("LAT_RENAL_DYSFUNCTION", "OUT_MORTALITY"),

    ("LAT_HEPATIC_COAG_DYSFUNCTION", "OUT_MORTALITY"),
    ("LAT_NEUROLOGIC_DYSFUNCTION", "OUT_MORTALITY"),
    ("LAT_METABOLIC_DERANGEMENT", "OUT_MORTALITY"),

    ("LAT_SHOCK", "OBS_BLOOD_PRESSURE"),
    ("LAT_SHOCK", "OBS_LACTATE"),
    ("LAT_RESPIRATORY_FAILURE", "OBS_OXYGENATION"),
    ("LAT_RENAL_DYSFUNCTION", "OBS_CREATININE_BUN_URINE"),
    ("LAT_HEPATIC_COAG_DYSFUNCTION", "OBS_BILIRUBIN_PLATELETS_INR"),
    ("LAT_NEUROLOGIC_DYSFUNCTION", "OBS_GCS_RASS"),
    ("LAT_METABOLIC_DERANGEMENT", "OBS_PH_ELECTROLYTES_GLUCOSE"),
    ("LAT_INFLAMMATION_SEPSIS", "OBS_TEMP_WBC_CULTURES"),
    ("LAT_CARDIAC_STRAIN", "OBS_TROPONIN_ECG"),

    ("TRT_ANTIBIOTICS", "OBS_CULTURES_MEDICATIONS"),
    ("TRT_VASOPRESSORS", "OBS_BLOOD_PRESSURE"),
    ("TRT_MECH_VENT", "OBS_OXYGENATION"),
    ("TRT_MECH_VENT", "OBS_VENTILATOR_SETTINGS"),
    ("TRT_DIALYSIS", "OBS_CREATININE_BUN_ELECTROLYTES"),

    ("MISS_MEASUREMENT_INTENSITY", "OBS_AVAILABILITY"),
    ("MISS_MEASUREMENT_INTENSITY", "OBS_LAB_COUNTS"),
    ("MISS_MEASUREMENT_INTENSITY", "OBS_VITAL_COUNTS"),
]

```

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11. Decision-tree implementation specification

11.1 Required input columns

At minimum, generate one analytic row per ICUSTAY_ID and early window.

Column group	Required columns / features
Identifiers	subject_id , hadm_id , icustay_id , window_start , window_end
Background	age , gender , ethnicity , insurance , language , admission_type , admission_location , first_careunit
ICD-derived	comorbidity_count , elixhauser_score , chronic organ disease flags, malignancy flag
Vitals	HR, SBP, DBP, MAP, RR, SpO2, temperature
Neuro	GCS total/components, RASS, pupil response if available
Respiratory	PaO2, FiO2, PaO2/FiO2, PaCO2, pH, PEEP, ventilation status
Renal/output	creatinine, BUN, urine output, dialysis/CRRT flag
Metabolic	bicarbonate, base excess, lactate, sodium, potassium, glucose
Hepatic/coag	bilirubin, AST, ALT, albumin, platelets, INR, PT, PTT
Infection	WBC, cultures ordered/positive, antibiotic started
Cardiac	troponin, CK-MB, arrhythmia flag, ECG-report-derived flag if used
Treatments	vasopressor present, mechanical ventilation, dialysis, antibiotics, fluids
Missingness	count per variable, missing flag per variable, time since last observation

11.2 Derived summary features

For each numeric time-varying variable within the early window:

```
first_value
last_value
min_value
max_value
mean_value
slope_per_hour
std_or_variability
count_observed
missing_indicator
time_to_first_measurement
```



For irregular data:

- Use timestamps, not row order.
- Convert units before thresholds.
- Compute worst values clinically:
 - BP/SpO2/GCS/pH/bicarbonate: lower can be worse.
 - HR/RR/temp/glucose/sodium/potassium: both extremes can be worse.
 - creatinine/BUN/lactate/bilirubin/INR: higher worse.
- Compute PaO2/FiO2 only when paired within a clinically reasonable interval.



11.3 Expected output columns

```
LAT_CHRONIC_BURDEN
LAT_INFLAMMATION_SEPSIS
LAT_GLOBAL_SEVERITY
LAT_SHOCK
LAT_RESPIRATORY_FAILURE
LAT_RENAL_DYSFUNCTION
LAT_HEPATIC_COAG_DYSFUNCTION
LAT_NEUROLOGIC_DYSFUNCTION
LAT_METABOLIC_DERANGEMENT
LAT_CARDIAC_STRAIN
```

Optional uncertainty flags:



LAT_*_UNKNOWN
LAT_*_LOW_CONFIDENCE
MISS_MEASUREMENT_INTENSITY_SCORE

11.4 Threshold table

Feature	Threshold
Age burden	≥75 years
MAP low	<65 mmHg, or sustained <70 mmHg
SBP low	≤90 mmHg; ≤100 as weaker qSOFA-style threshold
Lactate high	>2 mmol/L
SpO2 low	<92%; severe <90%
PaO2/FiO2 abnormal	≤300
RR high	≥30; sepsis screening weaker threshold ≥22
GCS abnormal	<15; moderate ≤13; severe ≤8
Creatinine high	≥2.0 mg/dL or rise ≥0.3 mg/dL
Urine output low	<0.5 mL/kg/h over 6h or <500 mL/day
BUN high	≥40 mg/dL
Potassium abnormal	<3.0 or ≥5.5 mmol/L
Sodium abnormal	<130 or >150 mmol/L
Bicarbonate low	<18 mmol/L
pH abnormal	<7.30 or >7.55
WBC abnormal	<4 or >12 K/μL
Temperature abnormal	<36°C or ≥38.3°C
Bilirubin high	≥2 mg/dL
Platelets low	<100 K/μL
INR high	≥1.5
Glucose abnormal	<70 or >250 mg/dL

11.5 Validation checks

</> Python

📄

▶ Run

```
required_outputs = [
    "LAT_CHRONIC_BURDEN",
    "LAT_INFLAMMATION_SEPSIS",
    "LAT_GLOBAL_SEVERITY",
    "LAT_SHOCK",
    "LAT_RESPIRATORY_FAILURE",
    "LAT_RENAL_DYSFUNCTION",
    "LAT_HEPATIC_COAG_DYSFUNCTION",
    "LAT_NEUROLOGIC_DYSFUNCTION",
    "LAT_METABOLIC_DERANGEMENT",
    "LAT_CARDIAC_STRAIN",
]

for col in required_outputs:
    assert col in latent_tags.columns
    assert set(latent_tags[col].dropna().unique()).issubset({0, 1})
```

```
for forbidden in [
    "deathtime", "hospital_expire_flag", "expire_flag",
    "discharge_location", "los", "survival_time"
]:
    assert forbidden not in features_used_for_latent_tags

assert all(feature_time <= window_end for feature_time in extracted_event_times)
assert units_normalized is True
assert itemid_mapping_version_logged is True
assert missingness_flags_created is True
```

12. Rejected alternatives

Rejected item	Reason
Raw lab-to-lab causal graph	Labs are mostly manifestations of physiology, not primary causal mechanisms.
Direct edge from creatinine to mortality	Creatinine is a renal dysfunction proxy; renal dysfunction causes mortality risk.
Direct edge from lactate to mortality as primary cause	Lactate is mainly a hypoperfusion/metabolic stress marker.
Treating antibiotics as infection itself	Antibiotics are clinician response and suspected-infection proxy.
Treating vasopressors as shock itself	Vasopressors are treatment; shock is latent hemodynamic failure.
Using discharge location in latent tags	Post-baseline and outcome-adjacent leakage.
Using ICU length of stay in severity tags	Affected by survival, complications, and discharge practices.
Using hospital mortality to tune thresholds	Outcome leakage.
Fluid-balance burden as approved latent	Source-system differences and strong treatment dependence reduce identifiability.
Nutritional burden as approved latent	Albumin/weight/nutrition inputs are nonspecific and incompletely measured.
Separate hepatic and coagulation latents	Weak early support and overlapping SOFA-style organ dysfunction; merged.
Measurement frequency as biological severity	It is an observation process, not a physiologic state.
Conditioning on lab availability in causal mortality models	Can open collider paths through clinician suspicion and severity.
Dense edges among all organ dysfunctions	Overfits clinical correlations and obscures causal mechanisms.
Bidirectional acute-state feedback in a single static DAG	Real ICU physiology is dynamic; this static DAG must remain acyclic.

13. Final confidence and limitations

The strongest latent variables are:

Strongest variable	Why
LAT_SHOCK	Multiple strong indicators: BP, vasopressors, lactate, urine output, acidosis.
LAT_RESPIRATORY_FAILURE	Supported by SpO2, ABG, FiO2, PEEP, ventilation.
LAT_RENAL_DYSFUNCTION	Supported by creatinine, BUN, urine output, dialysis; KDIGO-compatible.
LAT_GLOBAL_SEVERITY	Multi-domain construct aligned with SOFA/OASIS-style severity.

Strongest variable	Why
LAT_METABOLIC_DERANGEMENT	Broad lab support and clear clinical mechanisms.

The weakest latent variables are:

Weakest variable	Why
LAT_CARDIAC_STRAIN	Troponin/ECG testing is selective; assay differences and renal confounding.
LAT_HEPATIC_COAG_DYSFUNCTION	Bilirubin/INR may be missing; anticoagulation and chronic liver disease confound.
LAT_INFLAMMATION_SEPSIS	Infection is partly hidden; cultures/antibiotics reflect clinician suspicion.
LAT_NEUROLOGIC_DYSFUNCTION	GCS/RASS are confounded by sedation and intubation.

The most reliable decision trees are LAT_SHOCK , LAT_RESPIRATORY_FAILURE , LAT_RENAL_DYSFUNCTION , LAT_METABOLIC_DERANGEMENT , and LAT_GLOBAL_SEVERITY . The most uncertain are LAT_CARDIAC_STRAIN , LAT_HEPATIC_COAG_DYSFUNCTION , and LAT_INFLAMMATION_SEPSIS .

The most uncertain DAG edges are:

Edge	Uncertainty
LAT_CARDIAC_STRAIN → LAT_SHOCK	True in cardiogenic shock, but cardiac biomarkers are selectively measured.
LAT_GLOBAL_SEVERITY → LAT_RESPIRATORY_FAILURE	Respiratory failure may also precede global severity. Static DAG simplifies dynamics.
LAT_INFLAMMATION_SEPSIS → LAT_HEPATIC_COAG_DYSFUNCTION	Plausible but not always identifiable without infection confirmation.
BG_ETHNICITY_INSURANCE_LANGUAGE → OUT_MORTALITY	May reflect social, structural, access, and measurement pathways; causal interpretation requires care.

Additional data that would improve identifiability:

- pre-ICU baseline labs,
- clinician-entered problem onset times,
- medication indication,
- reliable sedation depth and timing,
- radiology-derived respiratory findings,
- validated infection adjudication,
- echocardiography/ECG structured findings,
- baseline functional status and frailty,
- external validation dataset.

This graph is clinically plausible because it respects the ICU data-generating process: background risk and acute insults generate latent physiologic states; latent states generate measurements, treatments, missingness patterns, and mortality. It is not a statistically proven ground-truth DAG. It should be treated as a transparent, clinically motivated structural hypothesis for latent-tag generation, causal adjustment planning, sensitivity analysis, and mortality modeling.

